LEADING INNOVATION
Drones at Garuda Aerospace, Chennai



Photograph by BANDEEP SINGH



> **"The major disruption will happen in the delivery of medicines and emergency supplies.... If we are able to provide logistics services in rural India, people will get remote access to a doctor in the nearest town"**
>
> – **AGNISHWAR JAYAPRAKASH,** *Founder, Garuda Aerospace*

WHY IT IS A GAME CHANGER

In India, a number of trials have been conducted by various players—food delivery start-up Swiggy, Tata 1mg for medicine delivery and IndiaPost among others. In fact, the disruption caused by the Covid-19 pandemic proved to be a test bed for drone applications such as the delivery of medicines and vaccines to remote locations in the country during lockdowns.

Drones too have come a long way. A decade ago, drones for civilian applications were expected to remain airborne for a maximum of six minutes before the battery ran out. Now, the basic norm is 45 minutes. Drones that are deployed for surveying or mapping can stay up for two hours. That makes delivery drones the next big thing.

THE CHANGE-MAKERS

DRONACHARYA AERIAL INNOVATIONS

- Founded by Prateek Srivastava in Pune in 2017, it was the first to deliver drugs via drones in Telangana in Sept. 2020 when Covid was raging. Last year, it went public with an IPO
- Primarily offers a drone pilot training programme; has a tie-up with robotics firm Gridbots to manufacture drones; partnered with start-up TSAW for delivery operations

TECH EAGLE

- Founded by Vikram Singh and Anshu Abhishek in a hostel garage at IIT Kanpur in 2015, it has partnered with various state governments to deliver medicines and vaccines
- In May 2022, it ferried a 3 kg parcel for IndiaPost across 46 km in Gujarat's Kutch; has also run pilots for Zomato and Swiggy

GARUDA AEROSPACE

- Rural India is the key focus for Chennai-based start-up; claims to have 5,000 bookings for agri-drones used for spraying crops
- Has inked MoUs with Naini Aerospace Ltd and Bharat Earth Movers Limited for manufacturing drones; has also carried out trials for Swiggy and the hospital chain Narayana Health

WHAT INDIA NEEDS TO MASTER

Today, India has among the most liberal policies in the world for drones—90 per cent of India's airspace has been opened up as a green zone for drone operations up to a height of 400 feet while there are production-linked incentives for the manufacture of these aerial vehicles. An Unmanned Aircraft System Traffic Management (UTM) is in the works. All this is aimed towards making India a global drone hub by 2030. It has already led to a boom in drone activity. The civil aviation ministry anticipates the domestic industry to reach a market size of Rs 12,000-15,000 crore by 2026.

With 65-100 per cent of the components being imported at present, there are several technologies to work on: propulsion systems, firmware, flight operation software, data processing software, and so on. "When you look at components, basically the motors, propellers, electronic speed controllers, autopilots, various power management systems and navigation systems, there are 10-plus start-ups working on each of these segments," says Smit Shah, president of the Drone Federation of India.

Flying drones 'Beyond the Visual Line of Sight' (BVLOS) requires a host of other infrastructure to be in place, the key areas being precision navigation, precision landing and building in different levels of redundancy in communication systems for safety. The first adopters for drone delivery will likely be the healthcare sector, with diagnostic labs ferrying pathological tests from remote locations to bigger centres. In the case of e-commerce cargo, drones will likely be deployed for mid-mile connectivity—essentially transferring payloads of 50-100 kg from one hub to another. "The idea," notes Shah, "should not be to replace the conventional logistics chain with drones. The approach has to be strategic in nature." ■